SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)
Constituent of Symbiosis International (Deemed University), Pune
(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of india)
॥वसुधैव कुटुम्बकम्॥ Re-Accredited by NAAC with 'A++' Grade

Subject: Programming with Java

ASSIGNMENT No :2

TITLE : Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Name: Arpan Gupta
PRN: 25070122046
Class: CSE A2
Batch: 2025-29

Assignment 2: Methods and Static Members

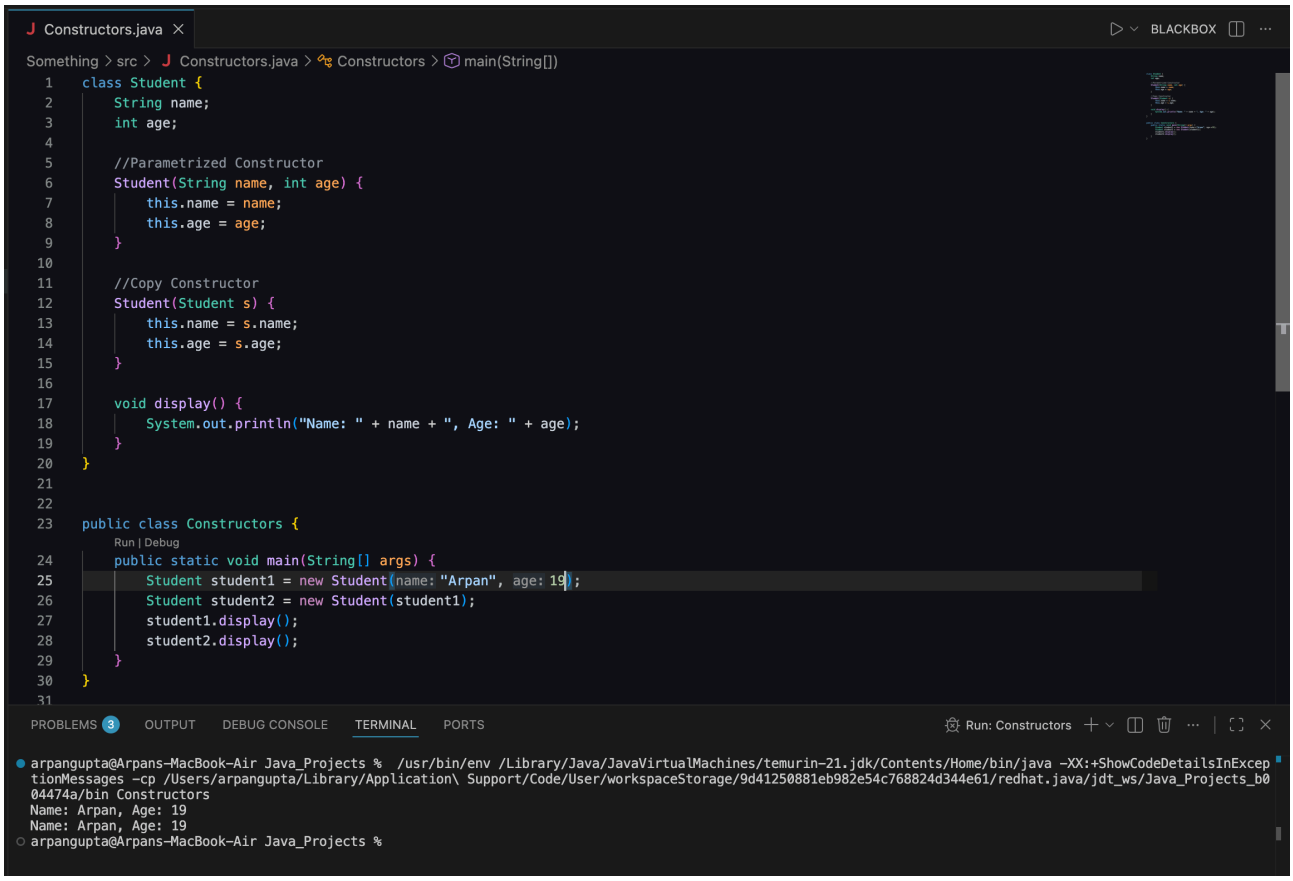
Problem Statement
Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Learning Objectives
To understand and implement default, parameterized, and copy constructors for object initialization.

Theory
Constructors are special member functions that are automatically invoked when an object is created. Java supports default constructors provided by the compiler, parameterized constructors for initializing objects with user-defined values, and copy constructors (implemented manually) for creating duplicate objects. Constructors improve code readability and simplify object initialization while supporting object-oriented programming principles. Constructors make object creation easy and efficient by assigning initial values to data members. They help avoid writing separate methods for initialization and ensure that every object starts with the required information. Different types of constructors provide flexibility in creating objects with default values, user-defined values, or copied values. As a result, constructors make Java programs more organized, readable, and easier to maintain.

Code and Output Screenshots

• Default Constructor



- Parameterized Constructor

Constructors.java

Something > src > Constructors.java > Constructors > main(String[])

```java
class Student {
    int age;

    //Parametrized Constructor
    Student(String n, int a) {
        name = n;
        age = a;
    }

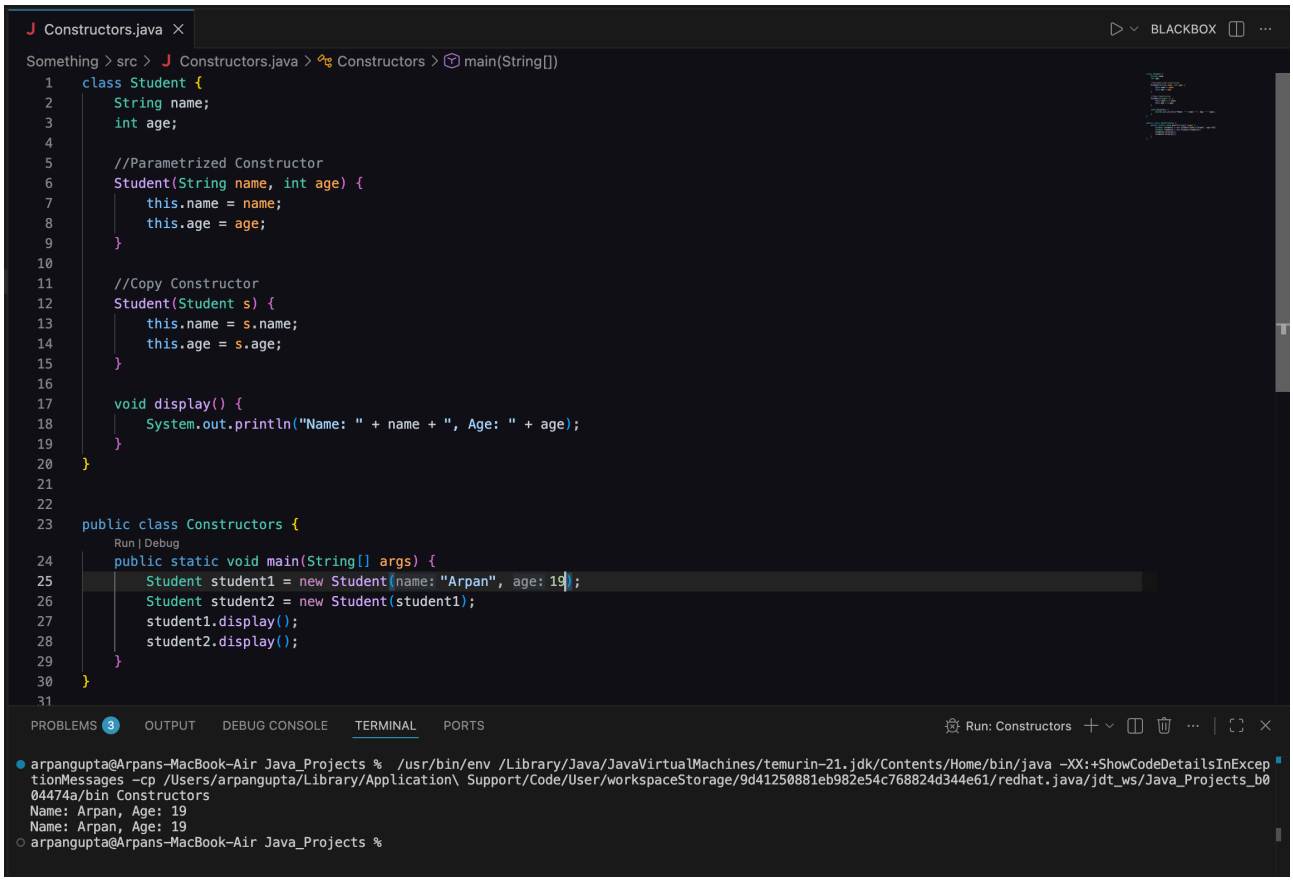
    void display() {
        System.out.println("Name: " + name + ", Age: " + age);
    }
}


public class Constructors {
    Run | Debug
    public static void main(String[] args) {
        Student student1 = new Student(n: "Arpan", a: 19);
        student1.display();
    }
}
```

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
arpangupta@Arpans-MacBook-Air Java_Projects %  cd /Users/arpangupta/Documents/Java_Projects ; /usr/bin/env /Library/Java/JavaVirtualMachines/temurin-21.jdk/Contents/Home/bin/java -XX:+ShowCodeDetailsInExceptionMessages -cp /Users/arpangupta/Library/Application\ Support/Code/User/workspaceStorage/9d41250881eb982e54c768824d344e61/redhat.java/jdt_ws/Java_Projects_b004474a/bin Constructors
Name: Arpan, Age: 19
arpangupta@Arpans-MacBook-Air Java_Projects %
```

• Copy Constructor

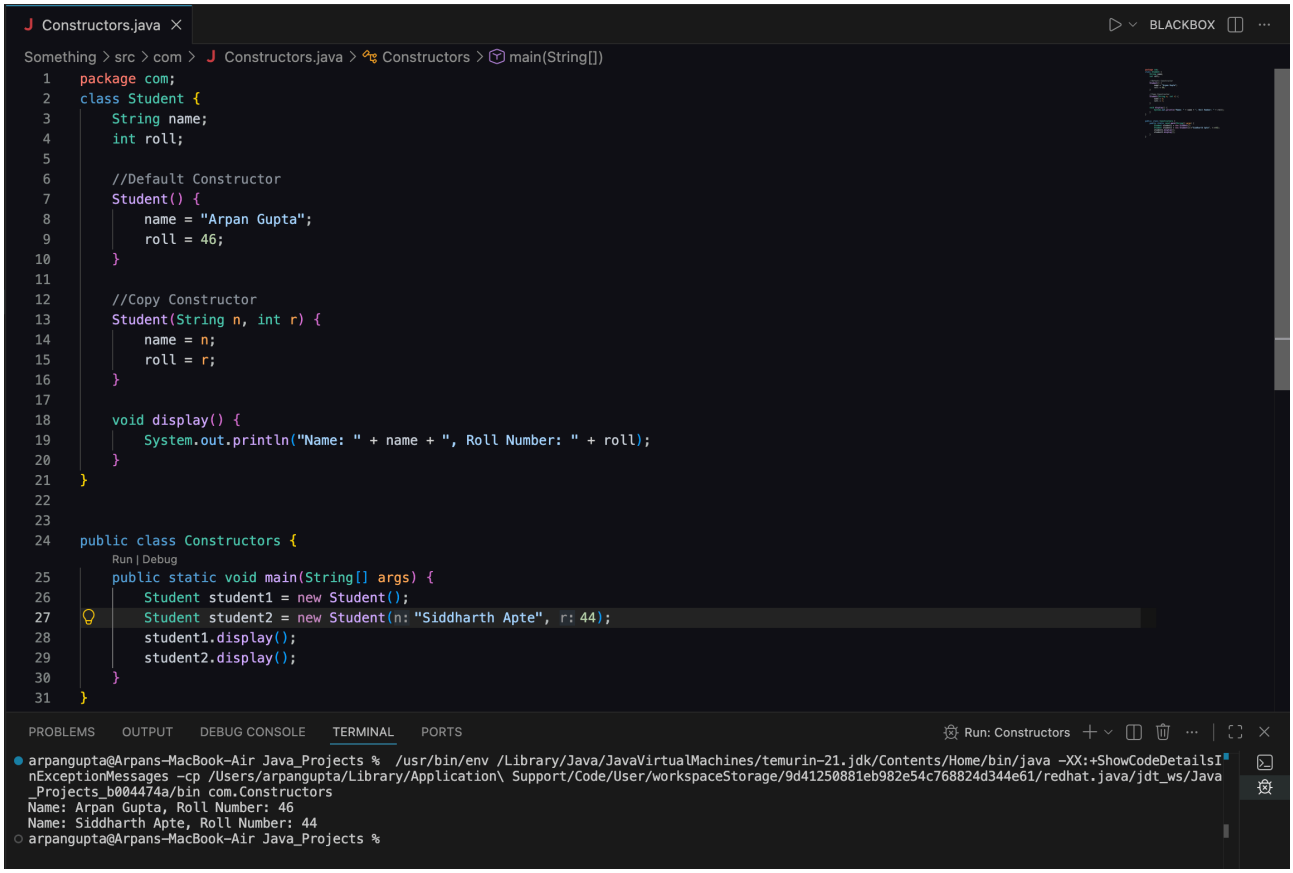


Conclusion

This assignment demonstrated the use of default, parameterized, and copy constructors in Java for object initialization. It helped in understanding how constructors simplify object creation, improve code organization, and support effective data management in object-oriented programming.

Exercise

1. Create a Student class using default and parameterized constructors to initialize student name and roll number.



2. Develop a Mobile Phone Inventory System using different constructors to initialize mobile details and create duplicate object records.

Construct.java

Something > src > lab > practice > Construct.java > Construct

```java
package lab.practice;
class Phone {
    String brand;
    String model;
    float cost;

    //Default Constructor
    Phone() {
        brand = "Apple";
        model = "iPhone 17";
        cost = 82900;
    }
    //Parametrized Constructor
    Phone(String b, String m, float c) {
        brand = b;
        model = m;
        cost = c;
    }

    //Copy Constructor
    Phone(Phone p) {
        this.brand = p.brand;
        this.model = p.model;
        this.cost = p.cost;
    }

    void display() {
        System.out.println("Brand: " + brand + ", Model: " + model + ", Cost: " + cost);
    }
}

public class Construct {
    Run | Debug
    public static void main(String[] args) {
        Phone phone1 = new Phone(b: "Samsung", m: "S26", c: 75999);
        Phone phone2 = new Phone(phone1);
        Phone phone3 = new Phone();

        phone1.display();
        phone2.display();
        phone3.display();
    }
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Run: Construct

```
arpangupta@Arpans-MacBook-Air Java_Projects %  /usr/bin/env /Library/Java/JavaVirtualMachines/temurin-21.jdk/Contents/Home/bin/java -XX:+ShowCodeDetailsInExceptionMessages -cp /Users/arpangupta/Library/Application\ Support/Code/User/workspaceStorage/9d41250881eb982e54c768824d344e61/redhat.java/jdt_ws/Java_Projects_b004474a/bin lab.practice.Construct
Brand: Samsung, Model: S26, Cost: 75999.0
Brand: Samsung, Model: S26, Cost: 75999.0
Brand: Apple, Model: iPhone 17, Cost: 82900.0
arpangupta@Arpans-MacBook-Air Java_Projects %
```